ESNet: A Dual-Modal Joint Framework of Elemental Composition and Crystal Structure for Material Properties Prediction

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Abstract

The study of predicting the properties of materials based on their structure is a crucial aspect in the field of materials science. Traditional methods usually focuses on studying the microscopic crystal structure of materials, failing to fully take into account the influence of the elemental composition of materials on their performance. In this work, we propose a dual-modal joint framework (called ESNet) . Specifically, ESNet builds on existing models based on crystal structure graph, to provide an in-depth analysis of how material elemental composition and crystal structure work together to influence material properties. To obtain the elemental composition information from the compound, we pose the elemental knowledge graph embedding technique to obtain the useful knowledge. To capture key information and potential connections between the two modes of elemental composition and crystal structure, we use a content-directed attention mechanism to dynamically focus on important regions in both features.It is worth noting that, despite its simplicity, ESNet outperforms existing methods in various material property prediction tasks on the Materials Project and Jarvis datasets.

1 Introduction

Material properties prediction is an important step in material engineering applications, such as discovering new materials with specific properties [Hamilton *et al.*, 2024] and evaluating the reliability of materials during their usage [Zhao *et al.*, 2024]. In the exploration of new materials and the optimization of existing materials, predicting material properties based on their structure is a core issue in materials science. The structure of materials refers to both the composition (atoms, molecules, ions) and the spatial arrangement formed by these components. Through this method, unnecessary experimental attempts can be minimized, and efforts can be focused on the most promising material systems and synthesis conditions based on the predicted results. This method thereby significantly enhances the efficiency of material research and development.

In recent years, high-throughput machine learning methods have been actively explored and applied in the research field of predicting material properties from crystal structures [Xie and Grossman, 2018; Schütt *et al.*, 2017; Choudhary and DeCost, 2021; Chen *et al.*, 2019; Louis *et al.*, 2020; Isayev *et al.*, 2017]. In this study, graph neural network encoders based on SE(3) invariance have been widely used to address the spatial invariance issue in crystal structures [Schmidt *et al.*, 2021; Reiser *et al.*, 2022]. Meanwhile, the method combining the Transformer encoder has also begun to demonstrate its potential. This method can achieve 3D rotation and translation equivariant attention mechanisms, fundamentally enhancing the accuracy of graph representation learning for crystal structures and the precision of property predictions. [Yan *et al.*, 2022].

The above studies of the crystal structure coding process mainly focus on the interaction between adjacent atoms, but the potential effect of material composition and atomic intrinsic physical properties (such as electronic structure, atomic radius, metallic properties, etc.) on properties is not fully considered. Previous methods [Wang *et al.*, 2021] have explored processing material elemental (an atom is the basic unit of an element) composition information using specific constraints or rules to uncover the relationship between composition and performance.However, these methods have not fully accounted for the inherent correlations among the three from the perspective of the material composition-structure-property relationship, leaving significant room for improvement.

In this work, we introduced a deep consideration of material elemental composition in the graph representation learning of crystal structures. A dual-modal joint framework ESNet (Elemental Composition and Crystal Structure Network) is proposed to comprehensively analyze and predict the material properties from composition and spatial structure, as shown in the Figure 1. In order to characterize the material elemental composition, we construct the element property knowledge graph. It is possible to preserve the intrinsic physical properties of the elements themselves, but also to capture the physical and chemical interactions and similarities between different elements. Specifically, ESNet consists of three modules: **(1) Elemental composition encoding module:** The atoms and their property nodes are encoded by knowledge graph embedding technology, and the features

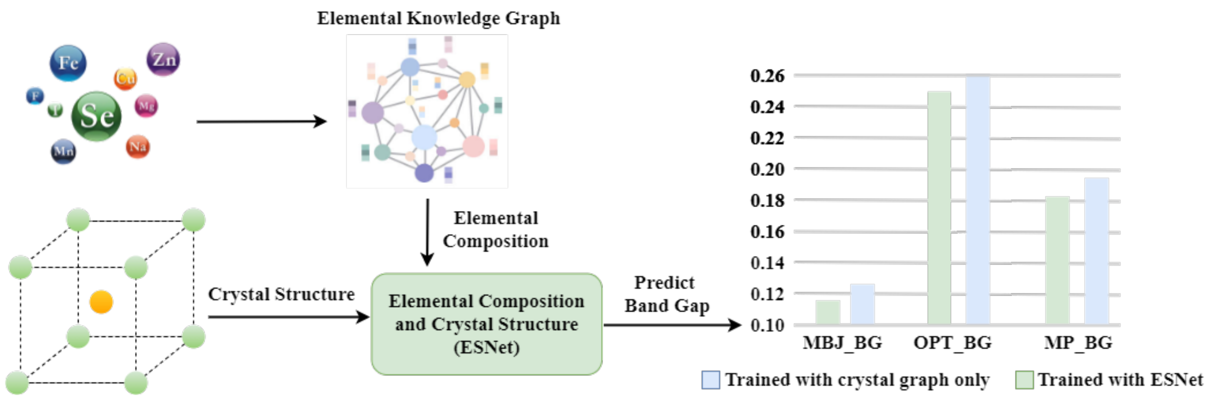

Figure 1: The proposed the dual-modal joint framework ESNet, which takes into account the material's elemental composition and crystal structure information to predict material properties. The figure shows MAE values for band gap prediction results.

of each atom and its property are extracted to form a high-dimensional and comprehensive component feature vector to represent the material composition information. **(2) Crystal structure encoding module:** Using a crystal graph designed with SE(3) invariance and SO(3) equivariance, this module integrates geometric information (such as Euclidean distance and angle information) to achieve efficient representation and processing of crystal structures. **(3) Feature fusion module:** By a feature generation process from coarse to fine, this module enables the deep fusion of material elemental composition features with crystal structure features, thus building a comprehensive representation of material properties. It is worth noting that to the best of our knowledge, our method is the first research to use the element property knowledge graph to enhance the prediction of crystal structure properties. Our contributions can be summarized as follows:

- In the form of knowledge graph, material elemental composition information is introduced to enhance the prediction method of material properties using only single mode of crystal structure.
- We propose a dual-modal joint framework ESNet for comprehensive analysis and prediction of material properties, which consists of elemental composition encoding module, crystal structure encoding module and feature fusion module.
- Performed extensive experiments on two benchmark datasets, Materials Project and Jarvis, validating the effectiveness and superiority of ESNet in predicting multiple critical material properties.

2 Related Works

Material Crystal Structure Representation. The crystal structure of a material is composed of the unit cell and the atomic motifs associated with the cell, and its mathematical expression is $M = (A, P, L)$ [Yan *et al.*, 2022].The crystal matrix $L = [l_1, l_2, l_3] \in \mathbb{R}^{3\times3}$ is used to describe the repeating pattern of the unit cell structure in three-dimensional space. Where $A = [a_1, a_2, ..., a_n] \in \mathbb{R}^{d\times n}$ contains the d-dimensional feature vectors of n atoms in the unit cell,

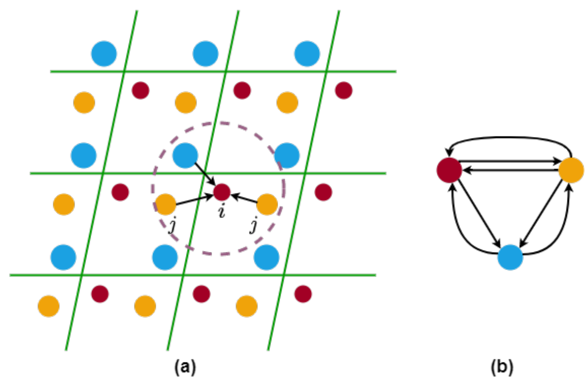

Figure 2: Illustrations of the crystal graph construction method(a) and the constructed graph(b).

$P = [p_1, p_2, ..., p_n] \in \mathbb{R}^{3\times n}$ contains the three-dimesional Cartesian coordinate of n atoms in the unit cell.In three-dimensional space, after infinite repetition, all the atoms in the unit cell can be expressed as:

$$p_n = p_i + k_1 l_1 + k_2 l_2 + k_3 l_3 \tag{1}$$

where three integers $(k_1, k_2, k_3) \in \mathbb{Z}$ define a 3D translation with (l_1, l_2, l_3). The crystal graph is constructed by building edges between atom j and atom i through a distance threshold r, i.e. $\|p_j + k_1 l_1 + k_2 l_2 + k_3 l_3 - p_i\|_2 = r$. In addition, due to the periodicity of the crystal, the same atoms may be connected in different cells, so there may be multiple edges between the atoms, forming a multi-edges graph. An example of a crystal graph construction is shown Figure 2). The purple circle (the radius is r) shows atom i's neighborhood, and all points in the circle have edges with atom i. Since atom j repeats twice within the domain of i, there are multiple sides from atom j to atom i. Formally, the crystals are 3D structures in practice, and we use illustrations in 2D for simplicity.

Crystal Graph Representation Learning. In the research of crystal structure encoding, several methods based on graph neural network(GNN) have been proposed and made significant progress. CGCNN [Xie and Grossman, 2018] represents

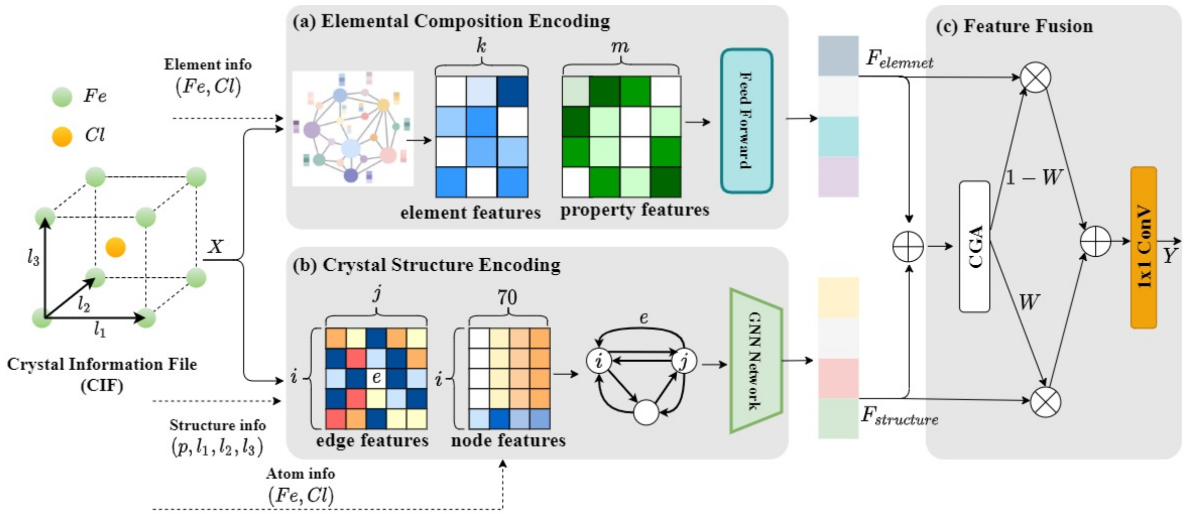

Figure 3: Overview of the proposed ESNet framework. Obtain elemental and structural information from the $FeCl$ Crystal Information File (CIF). **(a)**Elemental composition Encoding: Through knowledge graph embedding technology, it obtains the corresponding features of each element and the corresponding property features of each element according to the input element information (Fe, Cl). It then obtains the element component characteristic F_{element} through the network layer processing. **(b)**Crystal Structure Encoding: Based on the input structure information (where p represents the atomic position and (l_1, l_2, l_3) denotes the lattice matrix) and atomic information (Fe, Cl), the crystal graph is constructed. Subsequently, the crystal structure feature $F_{\text{structure}}$ is obtained through a Graph Neural Network (GNN). **(c)**Feature Fusion: Feature vectors F_{element} and $F_{\text{structure}}$ are fused through CGA.

the crystal structure graph, which lays the foundation for the application of GNN framework in crystal structure. In a similar approach, Schutt et al. [Schütt *et al.*, 2017] extended the molecular encoder SchNet to accommodate the crystal structure. Following this, a number of similar GNN methods have been proposed for general attribute prediction. For example, MEGNet [Chen *et al.*, 2019] introduced a method to progressively update atomic, bond, and global state properties through the GNN layer. GATGNN [Louis *et al.*, 2020] incorporated attention mechanisms into the convolution and pooling layers of the GNN framework. ALIGNN [Choudhary and DeCost, 2021] proposed a GNN for three-body interactions, enhancing model performance by utilizing atomic angle information. Recently, some studies have attempted to apply Transformer to crystal graph encoders. MatFormer [Yan *et al.*, 2022] focuses on solving the periodic encoding issue of crystal structures by designing a fully connected graph construction method to ensure the model is unaffected by periodic boundary changes. PotNet [Lin *et al.*, 2023] proposed a new type of physics-informed edge feature that embeds the infinite summation value of several known interatomic potentials and allows a standard fully-connected GNN to be informed of crystal periodicity. Comformer [Yan *et al.*, 2024] to achieve geometric completeness, introduces periodic vectors to construct lattice representations, ensuring that it satisfies the passive symmetry of crystals (including SE(3) invariance, SO(3) equivariance, and periodic invariance). It proposes two different methods: iComformer and eComformer.

Material Field Knowledge Graph Application. In the field of materials science, research on knowledge graphs is still in its infancy, and only a few researchers have explored the extraction of materials knowledge and conducted research on the extraction of generic types of knowledge such as material entities and relationships. In recent years, the construction and application of knowledge graphs in the field of materials is gradually becoming a research hotspot, and at the same time, significant results have been achieved. For example, Vineeth Venugopal et al. [Venugopal *et al.*, 2022] constructed MatKG, which is the largest knowledge map in materials science. Jamie P. McCusker et al. [McCusker *et al.*, 2020] proposed NanoMine, which is a knowledge graph focused on nanocomposite science. The correlation between microstructure and mechanical properties of 6XXX aluminum alloy [Zheng *et al.*, 2022] was studied based on high throughput calculation of knowledge graph. The KCL framework [Fang *et al.*, 2022] and the KANO framework [Fang *et al.*, 2023] apply the knowledge graph of chemical elements to molecular contrastive learning, which can leverage fundamental chemical knowledge in pre-training and fine-tuning to explore microscopic atomic interactions.

3 Methodology

Here, we propose the dual-modal joint framework ESNet. The framework accurately predicts various material properties by comprehensively analyzing their elemental composition and crystal structure. In particular, we obtain information about elements and crystal structures from the crystal information file(CIF). The overview of ESNet is inllustrated in Figure2, which consists of three main modules: (1) Elemental composition encoding module(Section3.1), as shown in . (2) Crystal structure encoding module(Section3.2). (3) Feature fusion module(Section3.3).

3.1 Elemental composition Encoding Module

Element Knowledge Graph Embedding. A typical representation method for material elemental composition is to transform the chemical information into composition-based feature vectors (CBFV [Kauwe *et al.*, 2018]). The CBFV method typically weights the contribution of elements by their abundance, which may overlook the complex interactions between dopant elements and the main elements. To resolve this issue, we construct an element knowledge graph to represent the relationships between elements and their properties. The element nodes in the knowledge graph can not only contain doping information but also capture the specific effects of dopant elements on physical properties such as electrical, mechanical, and thermal properties through attribute nodes, thus overcoming the limitations of the element vector combination approach in traditional CBFV methods. For continuous attributes, we discretize them into interval labels for relationship extraction. We obtained all elements and at least 15 of their properties(As shown in the figure 4) from the periodic table(https://ptable.com), forming 1,626 triplets that cover 227 entities (118 elements and 109 property intervals) and 17 types of relationships.

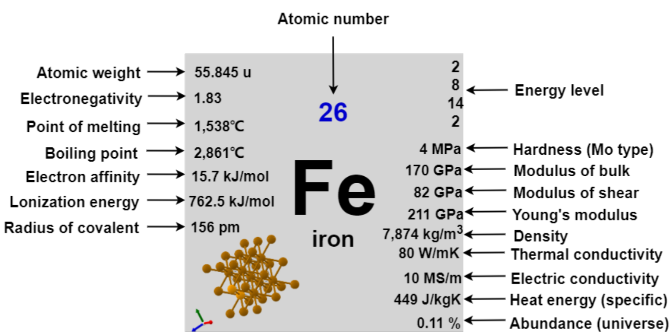

Figure 4: Illustrations of the property included in the Fe elements.

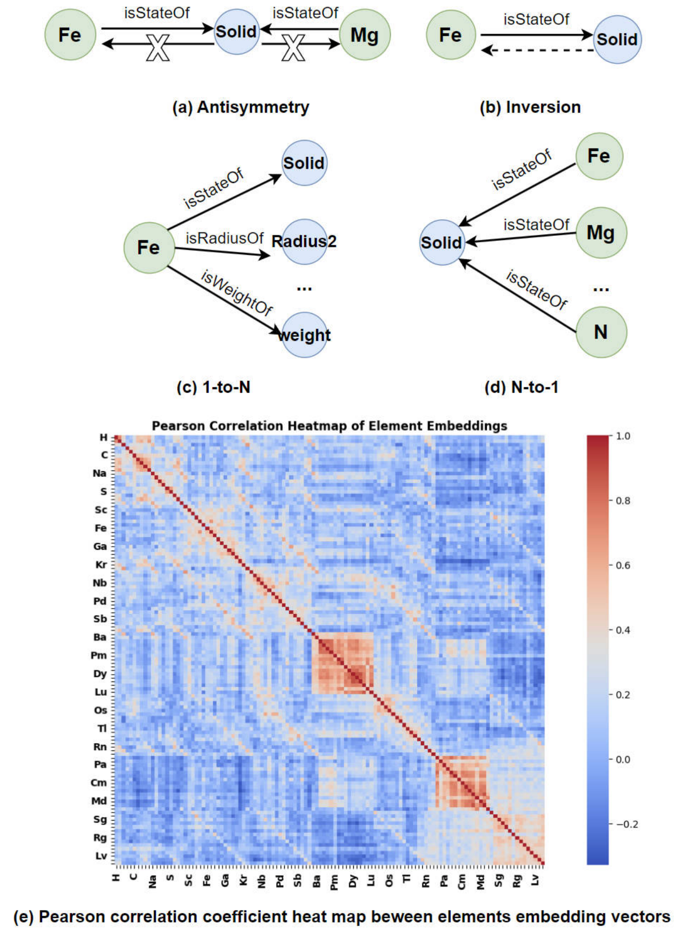

Figure 5: **a-d** Element knowledge graph relationship model. **(e)** Mapping attribute and Pearson correlation coefficient heat map

Considering the numerous relationship patterns (symmetrical, asymmetrical, inversion, and combination) and relationship mapping properties (1-to-1, 1-to-N, N-to-1, N-to-N) in the chemical element attribute triplets, as shown in Figures 5(a d), we employ the RotatE model [Sun *et al.*, 2019] for knowledge graph embedding (KGE) to better capture these intricate relationships and preserve the semantic similarities and differences between entities and relationships. We randomly sample a positive triplet (h_i, r_i, t_i) for each entity h from the knowledge graph. Then, we use the self-negative adversarial sampling method(Equation(2)) to generate the negative sample triplet $(h_j^{'}, r, t_j^{'})$ for (h_i, r_i, t_i).

$$p\left(h_j^{'}, r, t_j^{'} | (h_i, r_i, t_i)\right) = \frac{\exp \alpha f_r\left(h_j^{'}, t_j^{'}\right)}{\sum_i \exp \alpha f_r\left(h_i^{'}, t_i^{'}\right)} \tag{2}$$

where the variable α is defined as the sampling temperature. and $f_r\left(h_j^{'}, t_j^{'}\right)$ is the score of the triplet $(h_j^{'}, r, t_j^{'})$, represented by a distance metric(Equation(3)). Ultimately, the model learns more reasonable embeddings by maximizing the score function f_r for positive samples and minimizing the score function for negative samples.

$$L = -\log\sigma\left(\gamma - d_r\left(\mathbf{h}, \mathbf{t}\right)\right) - \sum_{i=1}^{n} p\left(h_i^{'}, r, t_i^{'}\right) \log\sigma\left(d_r\left(h_i^{'}, t_i^{'}\right) - \gamma\right) \tag{3}$$

To validate the effectiveness of the embeddings generated by the RotatE model, we extract the element embedding vectors and plot their Pearson correlation heatmap (Figure 5(e)) to examine whether the embeddings of the chemical element graph align with the trends of the periodic table. The Pearson correlation coefficient, which measures vector similarity, ranges from -1 to 1, where 1 indicates perfect positive correlation. In the heatmap, the diagonal elements exhibit a correlation of 1, reflecting self-correlation. The orange-red blocks near the diagonal indicate strong positive correlations between certain elements, which correspond to similar chemical properties. Elements within the same group (such as alkali metals and alkaline earth metals) display high correlation, consistent with the periodic trends.

Elemental composition Features. We innovatively apply knowledge graph embedding technology to deeply explore the features of elements and atomic intrinsic properties (attributes), thereby constructing an accurate and comprehensive

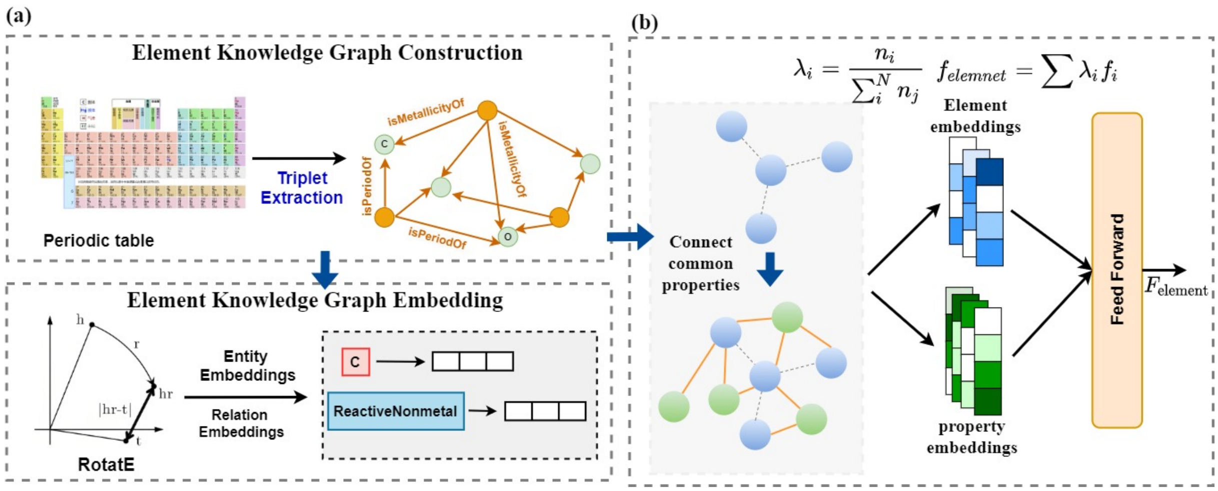

Figure 6: Illustrations of the material elemental composition encoding process. **(a)** Obtain elements and attributes from (https://ptable.com) to construct an element knowledge graph, and use RotatE for knowledge graph embedding technology to acquire entity and relationship encodings. **(b)** Search for the attributes (green dots) of all elements (blue dots) in the knowledge graph, filter the common attributes between elements. The formula $\lambda_i = \frac{n_i}{\sum_i^N n_j}$ is used to calculate the composition ratio of each element, where n_i represents the quantity of element i. N is the number of element types. Finally input both element features and attribute features into the network layer.

representation of material composition features(Figure 6). Through this technology, not only can the independent information of each element be fully preserved, but also, a bridge of information is built between elements through the chemical knowledge graph, enabling efficient sharing of information between elements. Specifically, for element feature extraction, we quantify the elemental composition by using the proportion of each element and perform a weighted processing of the individual element feature vectors. In feature extraction, we consider the commonalities between elements. By filtering the common properties of two elements and removing redundant data, we connect the two elements through common property nodes, further enhancing the expressive power of the features. Finally, we fuse the element features and property feature vectors through a linear layer in the deep neural network, obtaining the final material elemental composition feature representation, which provides richer and more accurate input features for downstream tasks.

3.2 Crystal Structure Encoding Module

Crystal Graph Initial. The initial node and edge features of our crystal graph encoder are shown in Table 1. The node features are one-hot encoded based on atomic property values, with a vector length of 70 dimensions. All atomic property values are calculated using the open-source package Mendeleev. The atomic properties used include (1) Lanthanide or Actinide, (2) Group number, (3) Perioid number, (4) Covalent radius, (5) Valence electrons, (6) First ionization energy and (7) Block. For discrete values, the vector is encoded according to the category to which the value belongs; for continuous values, the range of property values is divided into a specified number of categories, and the corresponding vector encoding is applied. Additionally, to distinguish between lanthanide and actinide elements, we apply special markers to the first three positions of the atomic feature vector. The edge features are represented by the spatial distances (Euclidean distances) between atoms and are embedded as initial edge features using a radial basis function (RBF) kernel.

Level	attributes name	#of categories
Edge	Atom distance	$c/\left\|e_{ij}\right\|_2$
	Lanthanide or Actinide	4
Node	Group number	18
(Atom)	Perioid number	9
	Covalent radius	10
	Valence electrons	17
	First ionization energy	10
	Block	4

Table 1: The node and edge features used in the graph encoder.

Crystal Structure Features. iComFormer [Yan *et al.*, 2024] demonstrates unique advantages in processing graph-structured data related to crystalline materials. It utilizes SE(3)-invariant crystal graph representations, which effectively capture the geometric information inherent in crystal structures. During the feature extraction process, iComFormer updates node features through message passing and aggregation among nodes. Its Node-wise transformer layer and Edge-wise transformer layer process geometric information from different perspectives. The Node-wise transformer layer, while passing messages, comprehensively considers the central node, adjacent nodes, and edge features. It achieves message passing and aggregation through the computation of queries, keys, and values, with a computational complexity of O(nk), ensuring efficiency while fully capturing the relationships between nodes. The Edge-wise transformer layer updates edge information using angle features and lattice vector edge features, further enhancing the model's ability to understand the geometric relationships within crystal structures. With these designs, iComFormer excels in tasks such as predicting the properties of crystal materials, providing robust graph encoding capabilities for our research and facilitating more accurate analysis and processing of crystal structure-related data(Figure 7).

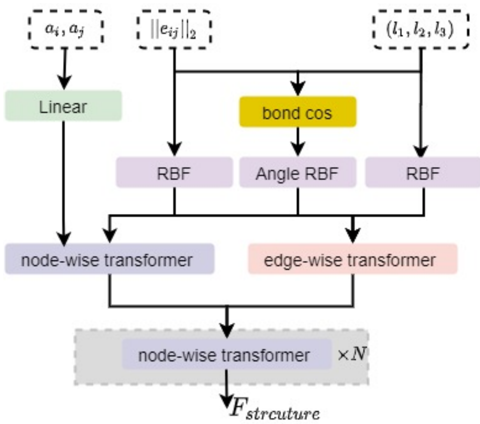

Figure 7: Illustrations of the processing flow of iComformer, where a_i, a_j are atomic features, $||e_{ij}||_2$ is edge features, and (l_1, l_2, l_3) is the lattice matrix. The number of layers in the node-wise transformer depends on the downstream task layers N. Specifically, $N = 3$ for calculating the bulk modulus and shear modulus, and $N = 4$ for other properties.

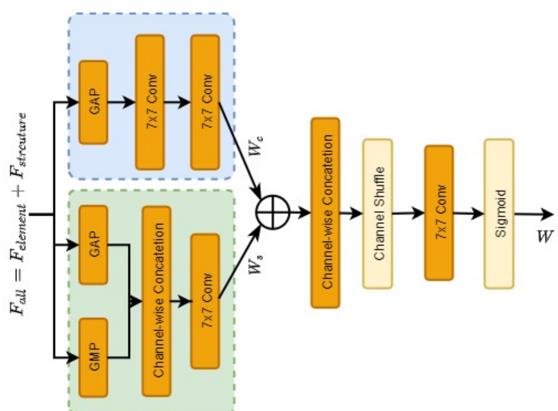

Figure 8: The diagram of content-guided attentine(CGA), where element composition features and crystal structure features are input simultaneously. CGA is a coarse-to-fine process: the coarse version of SIMs($W_{coa} \in \mathbb{R}^{C \times H \times W}$) is generated firstly and then every channel is refined by th guidance of input features.

3.3 Feature Fusion Module

In the feature fusion module, we employed the content-guided attention (CGA) [Chen *et al.*, 2024] feature fusion method, which enhances fusion accuracy and effectiveness by dynamically focusing on key regions in the input features. CGA employs a two-step attention mechanism, first generating coarse spatial attention features, then refining the fusion features of each channel through the input features. By combining spatial and channel attention, we reinforce the useful information in the features, dynamically adjust the weight of each channel, highlight important regions, and thus improve the feature fusion accuracy and the model's ability to understand complex material characteristics. The detailed procedures of CGA are illustrated in Figure8. We first compute the corresponding W_c and W_s by following:

$$W_c = \mathcal{C}_{1\times1}(\max(0, \mathcal{C}_{1\times1}(F_{all}{}^{c}_{(GAP)}))) \tag{4}$$

$$W_s = \mathcal{C}_{7\times7}([F_{all}{}^{s}_{(GAP)}, F_{all}{}^{s}_{(GMP)}]) \tag{5}$$

where $F_{all} = F_{element} + F_{structure}$ denote the combination of element composition features and crystal structure features. $max(0, x)$ denotes the ReLU activation function, $C_{k\times k}$ denotes a convolutional layer with kernel size $k \times k$, and $F_{all}{}^{c}_{GAP}$, $F_{all}{}^{s}_{GAP}$ and $F_{all}{}^{s}_{GMP}$ denote the features processed by the global average pooling operation across spatial dimensions, the global average pooling operation across channel dimensions, and the global maximum pooling operation across channel dimensions. Then we fuse W_c and W_s together via a simple addition operation, which follows broadcasting rules, to obtain the coarse SIMs $W_{coa} \in \mathbb{R}^{C\times H\times W}$. we experimentally find the product operation can achieve similar results.

$$W_{cos} = W_c + W_s \tag{6}$$

In order to obtain the final refined W, it is necessary to adapt each channel to the input features. The content of the input features is then used as a guide to generate the final channel-specific W. The final channel-specific W is generated using the content of the input features as a guide:

$$W = \sigma\left(\mathcal{GC}_{7\times7}\left(CS\left([F_{all}, W_{cos}]\right)\right)\right), \tag{7}$$

where σ denotes the sigmoid operation and $\mathcal{GC}_{k\times k}$ denotes the group convolutional layer with kernel size $k \times k$. In our implementation, the number of groups is set to C. Finally, we obtain the fused feature vector as:

$$Y = \mathcal{C}_{1\times1}(F_{structure} \cdot W + F_{element} \cdot (1 - W)) \tag{8}$$

4 Experiments

We evaluate the performance of ESNet on the widely-used Materials Project [Chen *et al.*, 2019] and Jarvis [Choudhary *et al.*, 2020] crystal materials benchmark dataset. Through experiments, our proposed ESNet demonstrates excellent performance in predicting material properties such as band gap and formation energy, validating the necessity of incorporating elemental attributes. We follow the experimental setting of ComFormer [Yan *et al.*, 2024] and use Mean Absolute Error (MAE) as the evaluation metric. Benchmark methods include CGCNN [Louis *et al.*, 2020], schNet [Schütt *et al.*, 2017], MEGNET [Chen *et al.*, 2019], GATGNN [Louis *et al.*, 2020], ALIGNN [Choudhary and DeCost, 2021], Matformer [Yan *et al.*, 2022], PotNet [Lin *et al.*, 2023], iComFormer [Yan *et al.*, 2024]. The the best experimental results highlighted in bold.

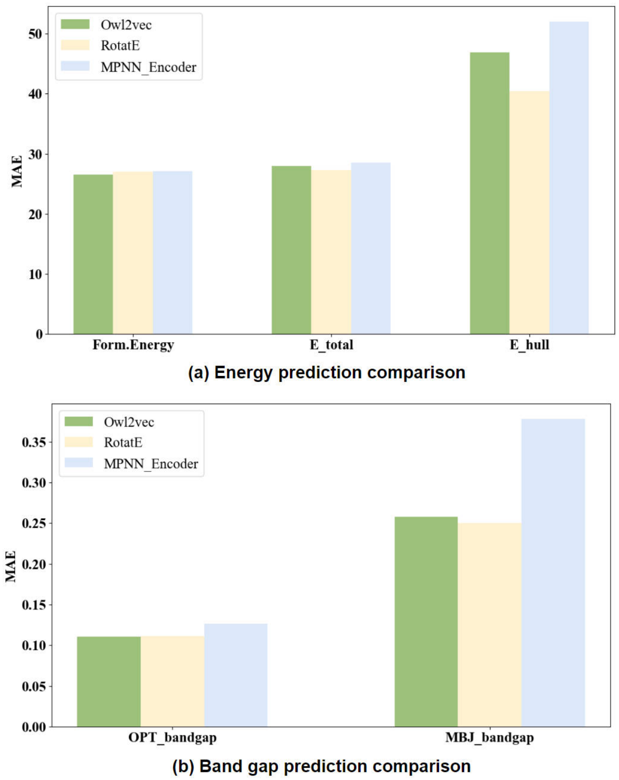

Figure 9: Comparative experiment on the effect of element knowledge graph Embedding.

4.1 Experimental results

Materials Project. The Materials Project-2018.6.1 was first proposed and used by MEGNET [Chen *et al.*, 2019] collected from The Materials Project [Persson, 2013], but the methods are compared by using different random seeds and dataset sizes. To make a fair comparison, all baseline methods using the same data splits. We follow the experimental settings and data splits of Matformer [Yan *et al.*, 2022] and PotNet [Lin *et al.*, 2023]. The formation energy and bandgap prediction tasks contain 60000, 5000, and 4239 crystals for training, evaluating, and testing. The bulk moduli and shear moduli prediction tasks contain 4664, 393, and 393 crystals for training, evaluating, and testingThe evaluation metric used is Mean Absolute Error (MAE). Experimental results on MP are shown in Table 2.

JARVIS. The JARVIS [Choudhary *et al.*, 2020] comprises multiple datasets, including JARVIS-DFT, JARVIS-FF, JARVIS-QETB, and JARVIS-ChemNLP, among others. These datasets encompass a broad spectrum of material domains, with materials processed through specific computational methods tailored to meet the requirements of various application fields. They offer robust data support and adaptability for a wide range of research and practical applications. The dataset utilized in this study is JARVIS-DFT, which contains over 80,000 materials computed using Density Functional Theory (DFT) and employs the OptB88vdW and TBmBJ methods, covering millions of properties.

We followed the methodology established by ALIGNN and utilized the same training, validation, and test sets to evaluate our ESNet on five significant crystal property tasks: formation energy, band gap (OPT), band gap (MBJ), total energy, and Ehull. The training, validation, and test sets comprise 44,578, 5,572, and 5,572 crystals for the formation energy, total energy, and band gap (OPT) tasks, respectively. For Ehull, the corresponding numbers are 44,296, 5,537, and 5,537, while for the band gap (MBJ), they are 14,537, 1,817, and 1,817. The evaluation metric employed is the Mean Absolute Error (MAE). The experimental results are shown in Table 3.

Method	Form.Enery	Band Gap	Bulk Moduli	Shear Moduli
	meV/atom	eV	Log(GPa)	Log(GPa)
CGCNN	31	0.292	0.047	0.077
schNet	33	0.345	0.066	0.099
MEGNET	30	0.307	0.060	0.099
GATGNN	33	0.280	0.045	0.075
ALIGNN	22	0.218	0.051	0.078
Matformer	21.0	0.211	0.043	0.073
PotNet	18.8	0.204	0.040	0.065
iComFormer	18.26	0.193	0.038	**0.0637**
ESNet	**17.76**	**0.1875**	**0.0369**	0.0659

Table 2: Comparison on The Materials Project.

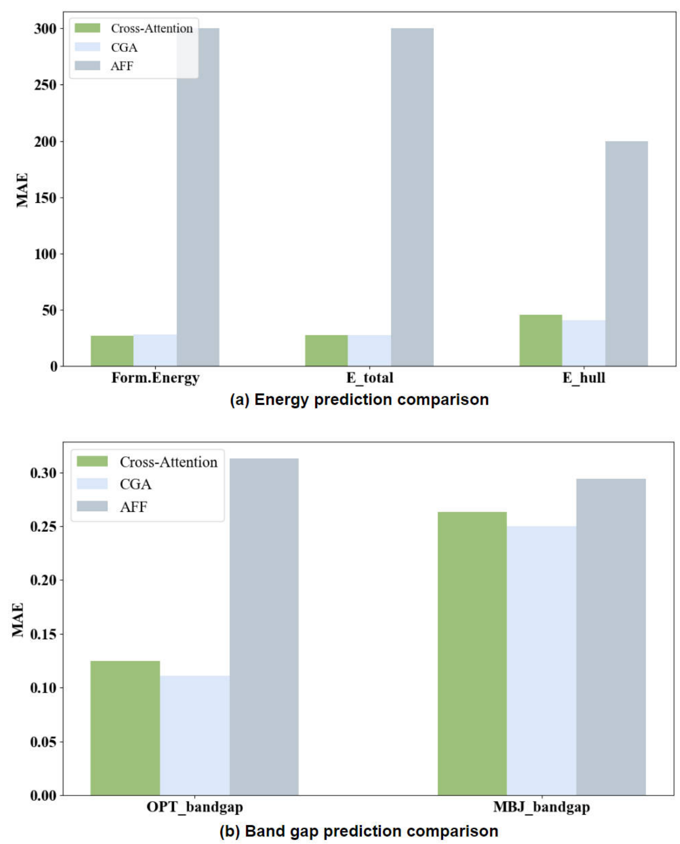

Figure 10: Comparative experiment on the effect of element knowledge graph Embedding.

Method	Form.Enery	Bandgap(OPT)	E_{total}	Bandgap(MBJ)	E_{hull}
	meV/atom	eV	meV/atom	eV	meV
CGCNN	63	0.20	78	0.41	170
schNet	45	0.19	47	0.43	140
MEGNET	47	0.145	58	0.34	84
GATGNN	47	0.170	56	0.51	120
ALIGNN	33.1	0.142	37	0.31	76
Matformer	32.5	0.137	35	0.30	64
PotNet	29.4	0.127	32	0.27	55
iComFormer	**27.2**	<u>0.122</u>	<u>28.8</u>	<u>0.26</u>	<u>47</u>
ESNet	<u>27.8</u>	**0.111**	**27.77**	**0.25**	**40.4**

Table 3: Comparison on The Javis.

4.2 Contrast experiment

In this study, two sets of comparative experiments were conducted on the Jarvis dataset. The first set of experiments was based on the Elemental Component Encoding Module, and the second set of experiments was based on the Multidimensional Feature Fusion Module. The purpose of conducting these experiments was to provide strong evidence of the effectiveness of the adopted approach through intuitive data.

The initial series of experiments is centred on a comparison of the effects of disparate Knowledge Graph Embedding methods. We have selected three commonly used knowledge graph Embedding methods, Owl2Vec [Chen *et al.*, 2021], RotatE, and knowledge graph MPNN encoder. As demonstrated in Figure 9, Owl2Vec and RotatE demonstrate comparable performance in bandgap prediction. However, in the prediction of the three energy properties, the efficacy of RotatE is more pronounced.Following a comprehensive evaluation, the decision was made to adopt RotatE.

The second set of experiments aims to compare the effectiveness of different multi-feature fusion methods.The RotatE model is uniformly used to generate Embedding for elemental knowledge graphs, and then Cross Attention, AFF (Attention Fusion Framework) and CGA (Cross-Graph Attention) are employed for feature fusion, respectively. The objective of the present study is to make a comprehensive evaluation of the performance of different fusion strategies in the scenarios under consideration, and thereby identify the most suitable feature fusion method for this study. As demonstrated in Figure 10, the CGA method demonstrates superiority over AFF and cross-attention in all property predictions, exhibiting a substantial enhancement in effectiveness.

5 Conclusion and Future Work

In this study, an innovative dual-modal joint framework(called ESNet). The framework accurately predicts various material properties by comprehensively analyzing their elemental composition and crystal structure. Despite the positive results of our research, there are still several research directions that deserve to be explored in depth based on the current findings. (1) The prediction results of shear modulus by the ESNet framework are unsatisfactory. Future research will concentrate on enhancing the prediction accuracy of these mechanical properties and refining the feature extraction and fusion strategies to capture the relevant physical laws. (2) The processing of non-periodic structures has not yet been addressed by ESNet. In the future, it is planned to extend the application scope of the framework to study new graph neural network models applicable to non-periodic structures, combining integrated modelling of element properties and structural features to provide efficient prediction and design support for complex structures.

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